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(no subject)

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<https://wokwi.com/projects/460666550942970881>
<https://wokwi.com/projects/460622961506022401>
<https://wokwi.com/projects/460622288796029953>
<https://wokwi.com/projects/460621112546152449>

host website

STEP 1: Launch EC2 Instance

1. Go to **AWS Console**
2. Open **EC2 Dashboard**
3. Click **Launch Instance**
4. Choose:
 - OS: **Amazon Linux 2** (recommended)
5. Instance type:
 - t2.micro (free tier)
6. Create/select **Key Pair (.pem file)** ⚠ (VERY IMPORTANT)
7. In **Security Group**:
 - Allow:
 - SSH (port 22)
 - HTTP (port 80)
 - Custom (like 3000 if needed)

👉 Click **Launch**

🔑 STEP 2: Connect from Local Machine (Terminal)

Go to folder where .pem file is stored.

Fix permission (IMPORTANT)

```
chmod 400 your-key.pem
```

Connect using SSH

```
ssh -i your-key.pem ec2-user@YOUR_PUBLIC_IP
```

👉 Example:

```
ssh -i mykey.pem ec2-user@13.233.xxx.xxx
```

📁 STEP 3: Upload Files from Local → EC2

Use **scp** command:

```
scp -i your-key.pem -r dist/* ec2-user@YOUR_PUBLIC_IP:/home/ec2-user/
```

👉 If full folder:

```
scp -i your-key.pem -r dist ec2-user@YOUR_PUBLIC_IP:/home/ec2-user/
```



STEP 4: Setup Server on EC2

After SSH login:

Update system

```
sudo yum update -y
```

Install Node.js (if needed)

```
curl -fsSL https://rpm.nodesource.com/setup\_18.x | sudo bash -  
sudo yum install -y nodejs
```

OR install nginx (for static site)

```
sudo yum install nginx -y  
sudo systemctl start nginx  
sudo systemctl enable nginx
```



STEP 5: Host Your Website

For static site (React/Vite build)

Move files:

```
sudo cp -r dist/* /usr/share/nginx/html/
```

Restart nginx:

```
sudo systemctl restart nginx
```



STEP 6: Access Website



Open browser:

http://YOUR_PUBLIC_IP

✓ Your site should be live 🎉



Common Errors (IMPORTANT)



Permission denied (publickey)

✓ Fix:

```
chmod 400 your-key.pem
```



Connection timeout

✓ Check:

- Security group allows **port 22**
- Correct IP

✖ Website not loading

✓ Check:

- Port 80 allowed
- nginx running

Viva Answer (Short)

👉 "We connect to AWS EC2 using SSH with a key pair, transfer files using SCP, and host the application using a web server like Nginx."

docker image

STEP 1: Launch EC2 & Install Docker

Connect to your EC2 (via SSH), then:

```
sudo yum update -y
sudo yum install docker -y
```

Start Docker:

```
sudo systemctl start docker
sudo systemctl enable docker
```

Give permission:

```
sudo usermod -aG docker ec2-user
```

👉 Logout and login again (important)

STEP 2: Create a Simple App (Example)

```
mkdir myapp
cd myapp
```

Create a file:

```
nano app.js
```

Paste:

```
console.log("Hello from Docker on AWS!");
```

STEP 3: Create Dockerfile

```
nano Dockerfile
```

Paste:

```
FROM node:18
WORKDIR /app
COPY . .
CMD ["node", "app.js"]
```



STEP 4: Build Docker Image

```
docker build -t myapp .
```

Check:

```
docker images
```



STEP 5: Run Docker Container

```
docker run myapp
```

👉 Output:

```
Hello from Docker on AWS!
```



STEP 6: Create ECR Repository

1. Go to AWS Console
 2. Open **Amazon Elastic Container Registry**
 3. Click **Create Repository**
 4. Name: myapp
-



STEP 7: Login to ECR

Run (AWS CLI required):

```
aws ecr get-login-password --region YOUR_REGION | docker login --username AWS --password-stdin  
YOUR\_ACCOUNT\_ID.dkr.ecr.YOUR\_REGION.amazonaws.com
```



STEP 8: Tag Docker Image

```
docker tag myapp:latest YOUR\_ACCOUNT\_ID.dkr.ecr.YOUR\_REGION.amazonaws.com/myapp:latest
```



STEP 9: Push Image to ECR

```
docker push YOUR\_ACCOUNT\_ID.dkr.ecr.YOUR\_REGION.amazonaws.com/myapp:latest
```

👉 Now your image is stored in AWS 🎉



Viva Answer (VERY IMPORTANT)

👉 "We create a Docker image using a Dockerfile, build it using docker build, and push it to AWS ECR, which acts as a container registry."



Common Errors

❌ **docker: permission denied**

```
sudo usermod -aG docker ec2-user
```

✖ aws command not found

sudo yum install aws-cli -y

✖ login failed

✓ Check:

- Region
 - Account ID
 - IAM permissions
-

🏁 Short Summary (Exam)

👉 Build → Tag → Push → Store in ECR